

João Jerónimo

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🔗 JoaoJeronimoHW

Summary

PhD Economist and Data Scientist with 4+ years of experience turning large-scale data into actionable decisions. Specialised in causal inference, experimentation (A/B testing, DiD, Synthetic Control), and predictive modelling. Proven track record building production-grade pipelines and structural simulations to forecast behaviour under changing incentives — applicable to pricing, product analytics, and policy measurement.

Work experience

AI Evaluation Expert

Remote

micro1

Feb 2026 – Present

- Designed domain-specific evaluation frameworks to assess LLM performance on economics and data science tasks, including rubric development for causal inference workflows, econometric method selection, and code quality for OpenAI.
- Developed synthetic datasets with controlled statistical properties (heteroskedasticity, endogeneity, panel structures) to benchmark model reasoning under realistic conditions
- Authored expert-level task specifications simulating senior economist instruction, used to evaluate model capability across regression, forecasting, and causal identification tasks

Data Scientist (Doctoral Researcher)

Edinburgh, UK

Heriot-Watt University

Sep 2021 – Nov 2025

- Showed standard wage decomposition methods overstate industry pay gaps, with bias corrections ranging from 12 to 38 basis points depending on wage floor exposure — using a 50M+ row employer-employee panel.
- Cut macro forecast error by 60% (RMSE) by benchmarking an LSTM pipeline against the incumbent ARIMA model on a UK government forecasting dataset.
- Mapped firm-to-firm worker flows across 357K firms into a queryable graph, enabling causal inference on subsets where standard estimators break down due to sparse mobility.
- Built a counterfactual simulation engine in Python to forecast worker responses to minimum wage changes, replacing a static policy model with a behavioural one.
- Communicated complex modelling results to non-specialist audiences, including conference presentations and undergraduate instruction.

Data Research Analyst

Lisbon, Portugal

University of Lisbon

Jan 2019 – Jun 2021

- Pre-processing and streamlining data across multiple formats (.csv, .json and HTML)
- Automation of data extraction processes through web scraping using Python, resulting in data preparation time reduction from 5-6 days to under 2 hours.
- Writing VBA scripts for simultaneous operations in large matrices, significantly speeding up data operations required for pre-processing.

Skills

Causal inference: Synthetic Control, Diff-in-Diff, IV, A/B Testing - applicable to product experimentation, policy evaluation, and marketing measurement.

Methods: Linear regression, forecasting, time series, Bayesian econometrics, non-linear regression.

Coding: R (Advanced), Python (Advanced), Stata, SQL, Git/Github (version control), DuckDB.

Independent technical projects

Iberian Electricity Market Data Pipeline

github.com/mibel 

- Built a production-style data pipeline independently from scratch to ingest, clean, and model hourly observations from OMIE, ENTSO-E, and Open-Meteo into a DuckDB warehouse and an analysis-ready time-series panel for causal analysis of the Iberian gas-price cap.
- Tools Used: Python (pandas, requests, entsoe-py), DuckDB, SQL, Open-Meteo API, Git/GitHub, Jupyter

Macroeconomic Forecasting Engine

github.io.com/sfc 

- Built LSTM pipeline, benchmarked against ARIMA to enhance forecasting accuracy of a UK government forecasting model by close to 60% (in terms of RMSE).
- Tools Used: Python (scikit-learn, TensorFlow)

Medium-term Inflation Predictor

github.com/ml 

- Wrote and calibrated ARIMA, VAR and Bayesian UC-SV models to predict quarterly inflation. Pseudo out-of-sample model evaluation, robust standard errors and comparative forecast table.
- Tools Used: Python (numpy, SciPy, pandas)

IV Bias Correction Algorithm

github.com/macro 

- Implemented a bias-correction estimator in R/Stata to solve endogeneity issues in time-series data. Based on Wang and Bellemare (2019).
- Tools Used: R, Stata

Education

Heriot-Watt University

Oct 2021 – Jan 2026

PhD in Economics

- Research areas: macroeconomics, labour economics, applied econometrics, data science. Award: UK Research and Innovation (UKRI) Scholarship (≈£60,000).

University of Minho

Sept 2017 – Jan 2020

MSc in Economics (summa cum laude)

- Honors: National Award for Academic Merit (Top ranked students in Portugal, 2019); "Best Student" scholarships for highest GPA in 2018.

University of Minho

Sept 2014 – Jun 2017

BA in International Relations (magna cum laude)

- Honors: "Best Student" scholarships for highest GPA in 2016.

Journal publications

Interactions between financial constraints and economic growth

Jul 2023

J. Jerónimo, A. Azevedo, P.C. Neves, M. Thompson

The North American Journal of Economics and Finance, vol. 67

[10.1016/j.najef.2023.101943](https://doi.org/10.1016/j.najef.2023.101943) 